Frederick Wang

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Education

University of Toronto – M.Sc. Applied Computing (AI Concentration) **September 2026 – Present**
Sample Coursework: Neural Networks, Deep Learning, Imitation Learning for Robotics, Computational Linguistics

University of British Columbia – B.Sc. Computer Science and Statistics **September 2021 – May 2026**
Awards: Dean's List (2022 – 2026), Trek Excellence Scholarship (2022), J. Fred Muir Scholarship (2022)

Sample Coursework: Machine Learning, Relational Databases, Data Structures and Algorithms, Stochastic Processes

Skills

Languages: Python, SQL, TypeScript/JavaScript, Java, R, C++, C

ML/AI: PyTorch, scikit-learn, Hugging Face, VAEs, Imitation Learning, Time-Series Foundation Models, Bayesian Modelling

LLMs & Retrieval: Claude API, MCP, RAG, Hybrid Search, Cross-Encoder Reranking, Real-time Voice/TTS

Data & Cloud: Spark, Databricks, Unity Catalog, Great Expectations, SQL Server, AWS (Bedrock, OpenSearch), GCP, Azure

Tools: Git, GitHub Actions, pytest, React, FastAPI, Next.js, CARLA, NVIDIA Cosmos

Work Experience

Data Engineer Intern – British Columbia Investment Management **May 2025 – December 2025**
- Won 1st in an internal hackathon with a schedule-optimization algorithm that improved scheduling efficiency by 25%
- Led production deployment of the algorithm, projected to unlock $20M in enterprise value for a portfolio company
- Engineered and deployed a production MCP server on Databricks Apps connecting Claude.ai to Unity Catalog, cutting company-wide ad hoc analysis time by 50%
- Added SQL Server Managed Instance support to open-source Great Expectations for Databricks data validation
- Architected a RAG knowledge base for natural-language LLM querying of unstructured enterprise data

Teaching Assistant – University of Toronto (CSC207) & UBC (CS and Statistics) **September 2024 – Present**
- Lead weekly UofT CSC207 tutorials and labs on Java, object-oriented design, and Git-based team development
- At UBC, led ML-in-R tutorials, mentored 100+ students across 4 terms, and co-facilitated 400-level database lectures on query optimization, concurrency control, and crash recovery

Data Analyst Intern – Providence Research **January 2024 – August 2024**
- Built supervised ML models in R predicting ankle-surgery outcomes from patient risk factors; comparative cohort analyses informed surgeons' choice between procedure types
- Replaced manually regenerated R reports with interactive Power BI dashboards and maintained research databases

Projects

Recoverability-Aware World Models for Safe Autonomous Driving **October 2026 – Present**
- Co-developing a recoverability-aware world-model objective in PyTorch that predicts whether a vehicle can still recover, distilled into a DAgger-trained policy to cut false "safe" judgments of prediction-only models
- Building a headless CARLA pipeline that mines unrecoverable scenes and generates counterfactual variants with NVIDIA Cosmos for recovery-labelled training data

Undercurrent: Commodity Forecasting with Unilever H3L Labs **September 2026 – Present**
- Benchmarking the Chronos-2 time-series foundation model (with covariates) against a hierarchical Bayesian driver model on commodity prices, combining the two via Bayesian Predictive Synthesis
- Scoring leakage-safe probabilistic forecasts on CRPS and calibration

Talkback: Interactive Voiced Audiobooks **October 2026 – Present**
- Built and deployed a React + FastAPI app that reads books with an expressive multi-voice cast and lets users interrupt to question characters by voice, answering only from passages already read so nothing is spoiled
- Integrated real-time speech with barge-in turn detection, emotion-tagged TTS, voice cloning and lip-synced avatars

Unsupervised Patient Stratification **January 2026 – April 2026**
- Trained VAE and GMVAE models in PyTorch, compressing 18,572 sparse lab and Chinese free-text features into a 32-d latent space; K-Means recovered 7 clinical phenotypes (e.g., liver fibrosis, hypothyroidism) with no labels